Complex Numbers in Python

BEGINNER LESSON | Qspiders | Python 3

Learning goals: Recognize complex-number notation and check its type.

1. What is a complex number?

A complex number has a real part and an imaginary part. Python uses `j` for the imaginary part and calls this type `complex`.

2. Create and use

In `2 + 3j`, the real part is 2 and the imaginary part is 3.

```
number = 2 + 3j
imaginary = 5j
print(number)     # Output: (2+3j)
print(imaginary)  # Output: 5j
```

3. Check the type

Use `type()` to find a value's data type.

```
print(type(number))  # Output: <class 'complex'>
```

4. Key points

- Write `3j` without a space between 3 and `j`.

- Python uses `j`, not `i`. Write `1j` when the imaginary part is one.

- `5j` has a real part of zero.

- Use no quotes: `"2 + 3j"` is text, not a complex number.

5. Try it yourself

Create `value` with the value `4 + 2j`. Print its value and its type.